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Designing Injection Strategies in a Heterogeneous Giant Carbonate Field, Offshore Abu Dhabi: Integrating Data-Driven Diagnostics with Engineering Insight

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Summary

This paper presents an integrated workflow to optimize water injection strategy in a large carbonate reservoir offshore Abu Dhabi, characterized by vertical heterogeneity and lateral variation. The field currently operates over 100 water injectors (around 140 historically, including inactive wells) and five gas injectors. The reservoir is subdivided into 16 geological layers and six radial engineering sectors. Injectors are categorized into three functional types based on position relative to reservoir contacts: down-dip water injector (DDWI), roll-up water injector (RUWI), and inner-ring water injector (IRWI). The study integrates three main approaches. First, deterministic simulation is used to highlight the physical impact of each injector group. Second, statistical analysis employs a 55- case Latin Hypercube design across 16 independent injection rate limits to capture nonlinear interactions and rank correlations between the rate limits and simulation results of reservoir pressure, oil production, GOR, and water cut. Third, engineering judgment incorporates production trends and bottomhole pressure history to interpret results in the context of field operations and surface constraints. Key findings from all three approaches are aligned. RUWIs contribute most to pressure support but require careful control to avoid early water breakthrough. IRWIs clearly reduce GOR and stabilize volume allocation among water injectors, although their pressure support effect remains limited at this early stage of deployment. DDWIs show limited added value under current total water injection constraints. Two representative field examples illustrate these dynamics. A southern sector highlights the limited effectiveness of DDWI. A northwest sector shows early GOR improvement with a RUWI-IRWI combination around a high-GOR producer. These cases help validate the broader recommendations. The outcome is a flexible strategy for injection allocation: shift of water away from less effective DDWI, use of RUWI for effective reservoir pressure support, and deployment of IRWI where

GOR control is needed. The framework supports ongoing surveillance and adjustment, aiming to balance pressure, avoid early breakthrough, and adapt when new field data becomes available.

Introduction

Water-injection strategy in carbonate reservoirs is inherently difficult. Compared with sandstones, carbonates typically exhibit stronger vertical heterogeneity and kv/kh contrast, and frequent fracture/vug systems. These features lower sweep efficiency, complicate conformance control, and make gas/water breakthrough timing less predictable. Practical design therefore must balance pressure maintenance with gas/water breakthrough risk while respecting facility constraints (i.e., gas/water injection capacity). Experience from the mature asset highlights three recurring needs. First, location of injectors relative to fluid contacts (i.e., down-dip, roll-up, and inner- ring positions) as well as rate play distinct roles in pressure support, sweep efficiency, and breakthrough gas management. Second, surveillance-driven control (e.g., Voidage Replacement Ratio (VRR)/Hall-type diagnostics, pressure/production trends, and streamline- based sweep-pattern reviews) is essential to detect channeling and reallocation of water before oil production losses become long-term issues (Brouwer et al. 2004, Alhuthali et al. 2010, Peters et al. 2010, Park et al. 2011). Third, regularly updating models and adjusting rates step by step tend to work better than making one-off design choices—especially as understanding of geology and operational efficiency evolve with additional drilling/surveillance dataset (Wang et al. 2009, Yousef et al. 2011). In this paper, "deterministic simulation" tries to clarify physical mechanisms contributed by the different injector category (i.e., down-dip water injectors (DDWI) below FWL; roll-up water injectors (RUWI) above FWL; inner-ring water injectors (IRWI) near gas-oil-contact (GOC)) employing simplistic extreme case settings: ON/OFF scenarios. Subsequently, statistical exploration tries to capture non-linear interactions among multiple variables and rank their correlations that are potentially overlooked in the simple deterministic simulation. Engineering judgment then reconciles models with field production profiles and observed pressure. The integration of the deterministic, statistical, and engineering judgment will bring to (i) identify where injected water adds value, (ii) determine how much to inject for each injector type, and (iii) specify how to adapt when surveillance diverges from plan. The objective of this work is to showcase a practical framework that improves confidence in day-to-day injection rate allocation while remaining room for adaptation as new field data is integrated.

Methodology

The methodology consists of three main approaches as summarized in Fig. 1: (1) Sensitivity Study with Extreme Value Range (On/Off sensitivity) as the deterministic simulation to understand physical mechanisms in responses of the different injector types, (2) Exploration of statistical correlation with Experimental Design as the statical simulation using Latin Hypercube Sampling (LHS) to extract trends across multiple variables, and (3) expert-led analytical evaluation based on field surveillance and insights as engineering judgement. These perspectives are integrated to guide a sector-wise injection strategy that reflects both subsurface complexity and operational constraints.

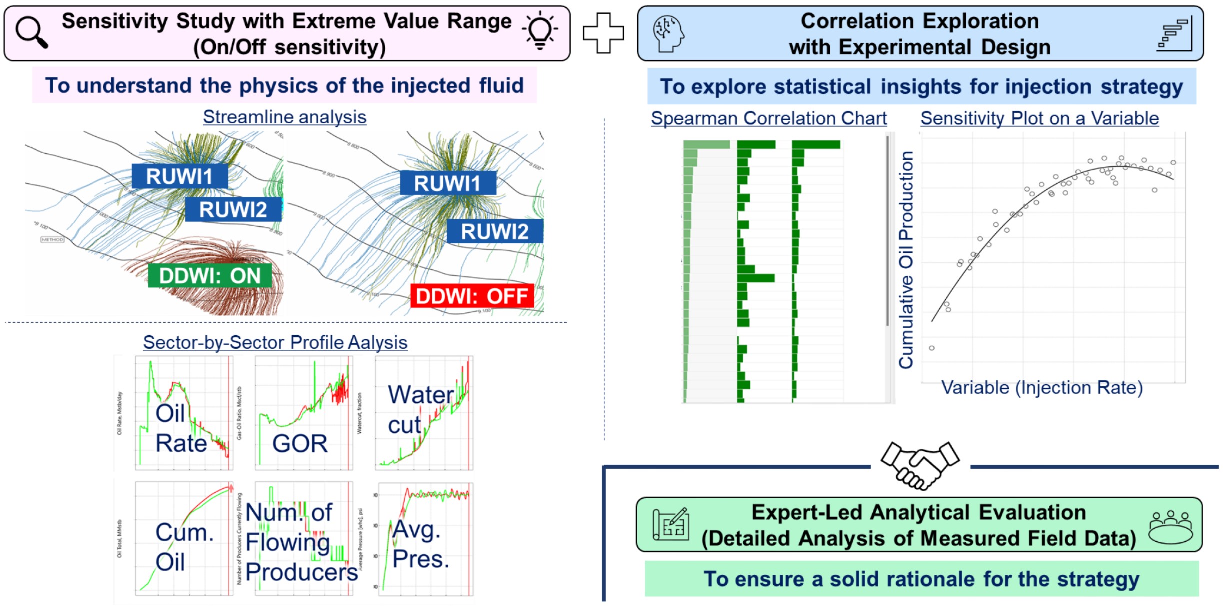

Figure 1—Integrated approach to optimize injection strategy through deterministic simulation, statistical correlation, and expert analysis.

The locations of the DDWI, RUWI, and IRWI, along with the sector divisions, are shown in Fig. 2. The reservoir is divided into radial sectors; these divisions are not based on geological features, but they have been artificially created for the purposes of easier reservoir management and engineering analysis. This artificial sectorization helps facilitate reservoir management for optimization and control. The field follows a concentric development scheme, where oil between the IRWIs and RUWIs/DDWIs is progressively recovered. Given the offshore nature of the field, it is challenging to drill all wells simultaneous; the injectors are drilled and implemented in a phased manner.

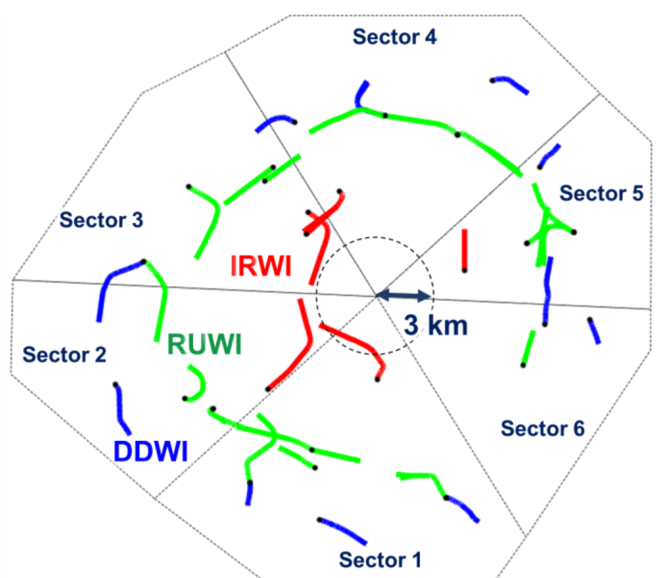

Figure 2—Locations of the down-dip water injector (DDWI), roll-up water injector (RUWI), and inner-ring water injector (IRWI), along with engineering sector divisions

Deterministic Simulation: Sensitivity Study with Extreme Value Range

To explore the physical role of each injector category, a series of ON/OFF sensitivity simulations was conducted. Three categories of water injectors were evaluated independently: DDWI, RUWI, and IRWI.

Gas injectors (GI) at crest were included in the base case and all runs as fixed elements. Four simulation cases were considered as shown in Table 1: the Base Case (all types of injectors are active), and three sensitivity cases (#A: No DDWI, #B: No RUWI, #C: No IRWI). Key simulation profiles cover sector-wise oil rate, GOR, water cut, cumulative oil, average sector pressure, and number of flowing producers. Streamline analysis was also performed to visualize sweep patterns and trace the behaviour of injected fluids under each scenario.

Table 1—Summary of deterministic simulation cases with extreme value settings (ON/OFF) for injection group.

SIMULATION CASES	DDWI	RUWI	IRWI	GI
BASE	Active	Active	Active	Active
#A (No DDWI)	Inactive	Active	Active	Active
#B (No RUWI)	Active	Inactive	Active	Active
#C (No IRWI)	Active	Active	Inactive	Active

Statistical Simulation: Correlation Exploration with Experimental Design

To further capture complex interactions across the reservoir, 55 simulation runs were performed using LHS. This allowed us to explore impacts of 16 continuous injection rate upper limits under reservoir-pressure-controlled constraints.

- DDWI: 5 variables (1-5,000 bbl/d/well across 5 sectors)
- RUWI: 6 variables (1-10,000 bbl/d/well across 6 sectors)
- IRWI: 5 variables (1-10,000 bbl/d/well across 5 sectors)

Although the maximum allowable injection rates were input as variables, actual injection volumes were dynamically adjusted by the simulator with the Proportional-Integral-Derivative (PID)-based pressure control. As a result, injection response was often non-linear and could not be interpreted solely from input rate magnitudes.

Two key outputs were analyzed:

- Spearman correlation coefficients between each variable and key simulation profiles (oil, pressure, GOR, and water cut)
- Variable-response sensitivity plots to observe the correlation and curvature of response (e.g., positive, plateauing, negative trends, or scattered)

This statistical approach complements the deterministic analysis by identifying variable combinations with the strongest statistical impact and revealing interactions that were not apparent in isolated ON/OFF tests.

Engineering Judgment: Expert-led Analytical Evaluation

In this study, engineering judgment functions as a standalone, co-equal method alongside deterministic and statistical simulations. This engineering judgement is an independent means of drawing field-based conclusions grounded in operational reality. The core of this engineering judgment lies in direct interpretation of field performance —specifically the relationships among production rates, injection volumes, and bottomhole closed-in pressure (BHCIP). These time-aligned signals are evaluated to identify patterns that may or may not be aligned with numerical forecasts. Rather than focusing on isolated events, this considers sustained operational behavior across relevant periods and within the broader injection strategy context.

This approach relies on experience-based pattern recognition; for example, assessing whether bottomhole pressure (BHP) stabilization follows a specific injector activation, or whether changes in GOR and water cut trends correspond to the start of water injection. Such assessments are performed at scale across the three injector types, DDWI/RUWI/IRWI, and reservoir zones to develop grounded conclusions about injector effectiveness and contribution. By embedding engineering judgment as an equal methodological pillar, this study ensures that decision-making is driven not solely by modeled expectations but by how the reservoir system actually behaves under injection. This approach increases robustness, adaptability, and field relevance in the development of long-term injection strategies.

Results and Discussion

This section is organized into four parts. First, the deterministic simulation examines the physical mechanisms of each water injector category using ON/OFF sensitivity. Second, the statistical exploration uses the LHS-based ensemble to capture non-linear interactions and rank their correlations. Third, the engineering judgment stands as a co-equal pillar, interpreting production, injection, and producer BHP histories to define workable operating envelopes. Finally, integration and summary consolidate the three lines of evidence into clear guidance on allocation and rate bands. Within each of the first three parts, we discuss injector categories in the same order— DDWI, RUWI, and IRWI—so that physics, data trends, and field signals can be compared side-by-side.

Deterministic Simulation: Sensitivity Study with Extreme Value Range

Down-Dip Water Injectors (DDWI).

Streamline Visualization - DDWI

To assess the flow behavior with and without DDWI, streamline visualization was conducted as shown in Fig. 3 using a representative sector, Sector 1, where two RUWIs (positioned slightly above the free water level) are paired with a DDWI. The streamline visualization highlights that, when the DDWI is active, the flow paths from the RUWIs tend to shift slightly upward and inward, reducing the amount of water migrating below the free water level. While the DDWI itself does not show clear flow paths reaching inward, its indirect effect contributes to guiding injected water from the RUWIs more effectively into the productive zone to some extent. This indirect pressure support, although not visually dominant in streamline density, sets the stage for further analysis via a tracer-based analysis.

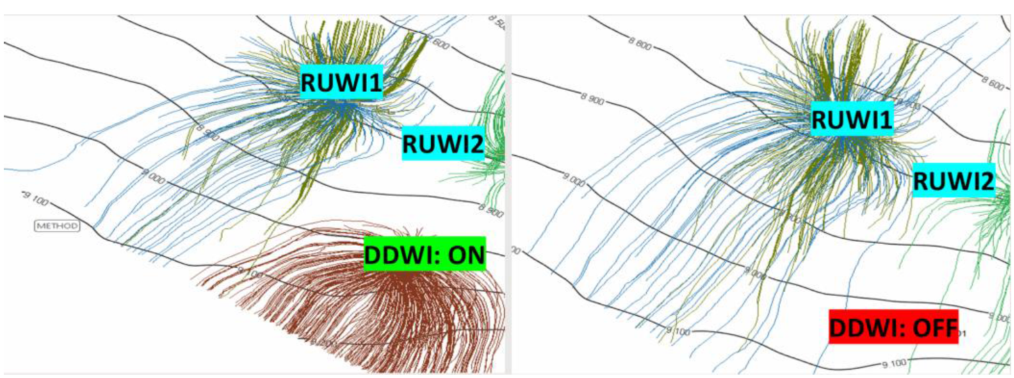

Figure 3—Streamline analysis comparing the impact of DDWI activation on flow dynamics in a selected sector. The left panel shows the streamline distribution with DDWI activated. The right panel illustrates the case with DDWI deactivated, where flow is solely by RUWIs. The difference highlights the supplemental pressure support provided by DDWI.

Tracer-Based Efficiency Assessment - DDWI

To quantify the directional effectiveness of roll-up water injection with and without down-dip support, tracer simulations were performed as illustrated in Fig. 4. A passive tracer was introduced into the RUWIs and its movement over time was tracked in simulation. The focus was on the fraction of the tracer reaching the zone above the free water level—defined as the oil-bearing region. Simulation results reveal that this upward-directed tracer fraction increased by approximately 2%point with the existence of the DDWI. While the difference is small, this method provides a practical, time-integrated view of sweep efficiency; tracer-based interpretation can be considered as a reliable proxy for understanding flow directionality over time in simulation environments.

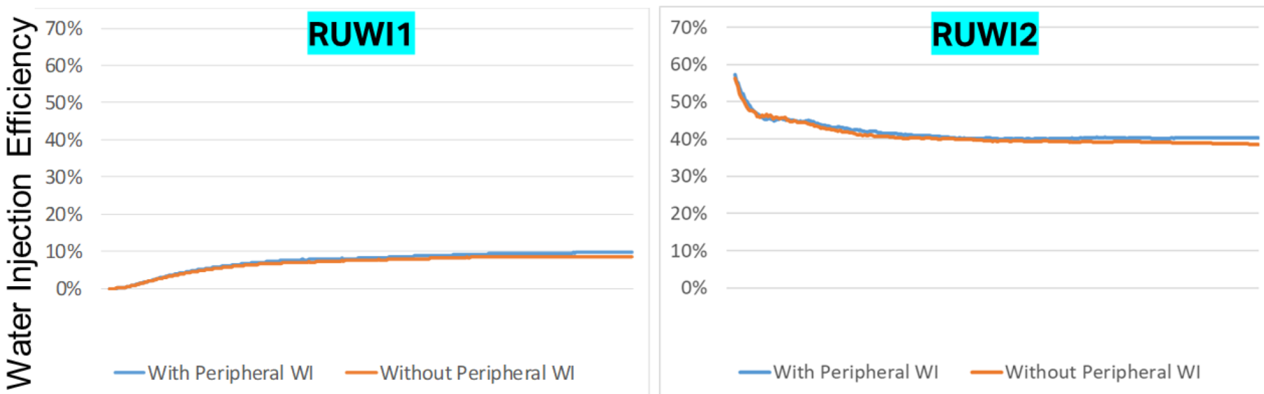

Figure 4—Water injection efficiency based on the ratio between injected tracer amount above free-water-level against total injected tracer amount. The left plot shows RUWI1 and the right is RUWI2. The subtle increment of efficiency by up to 2%pt can be observed in both plots.

Simulation profile - DDWI

Comparison of production and pressure profiles under the scenarios with and without DDWI are shown in Fig. 5. The oil rate profiles exhibit minimal difference in the early stage though the scenario without DDWI leads to slightly higher cumulative oil production in the later phase. This change seems to be caused by the total injection rate constraint applied to the field; larger injection-volume allocation to the down-dip area in the case with DDWIs reduces available injection volume to up-dip zones where pressure support to oil pool is more effective (i.e., RUWIs and IRWIs) leading to slower achievement of target sector pressure. The average reservoir pressure in this sector is expected to be higher by approximately 100 psi in the absence of DDWI, suggesting more effective pressure maintenance when water is injected directly into the oil zone. This result emphasizes the importance of injector placement and prioritization with consideration of operational constraints such as surface facility limits in the total injection volume capacity.

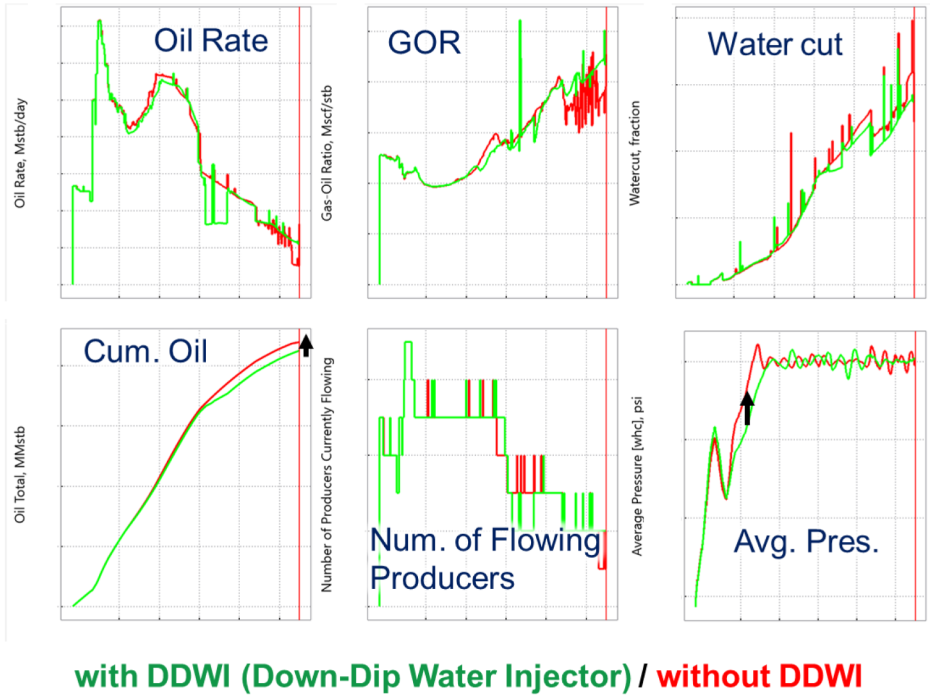

Figure 5—Sector-wise profile comparison of cases with and without DDWIs. Green curves represent the case with active DDWIs and red curves represent the case with inactive DDWIs. Cumulative oil shows a slight increase in the case without DDWI, accompanied by a higher average pressure caused by larger injection-volume-allocation to roll-up injectors leading to more effective pressure support to the sector.

Roll-Up Water Injectors (RUWI).

Pressure Map Comparison - RUWI

The impact of RUWIs on reservoir pressure support is evaluated through comparing pressure maps by cases with and without RUWIs five years after the initiation of injection in a representative sector, Sector 3 (Fig. 6). The differential pressure map exhibits predominantly blue-shaded regions, indicating a consistent increase in reservoir pressure at locations where RUWIs exist. This pattern suggests that RUWIs contribute significantly to pressure maintenance, especially in the oil-bearing zones located above the free-water- level. In several areas, the observed pressure increases by more than 500 psi, confirming effectiveness of RUWIs for pressure recovery in the nearby area. This indicates the RUWIs are particularly beneficial for enhancing pressure propagation toward oil-bearing zones, where pressure support from downdip side (i.e., DDWIs) may be limited.

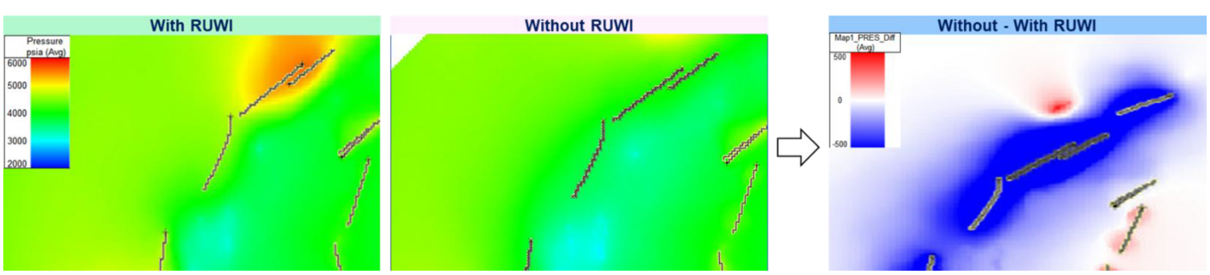

Figure 6—Pressure map comparison after five years of injection between the cases with and without RUWIs in a sector. The difference map shows pressure difference by subtracting the case with RUWIs from the case without RUWIs; the blue color indicates pressure increase by the RUWIs. The case with RUWI shows a distinct increase in reservoir pressure, indicating the significant contribution by RUWIs to pressure support in nearby area.

Simulation Profile - RUWI

Simulation profiles of Sector 3 demonstrate a pronounced performance difference associated with the presence of RUWIs (Fig. 7). Both oil rate and cumulative oil production indicate an improvement in recovery with the presence of RUWIs in the earlier stage. The presence of RUWIs enhances the average reservoir pressure within the sector by approximately 300 psi. Importantly, this pressure recovery is achieved without a substantial increase in water cut. Furthermore, GOR is better controlled under the case with RUWIs, indicating enhanced pressure support in high-GOR regions. These outcomes collectively confirm the effectiveness of RUWIs in maintaining reservoir pressure and enhancing recovery. These profiles, however, also reveal a nuanced operational challenge; the number of flowing producers is, for limited duration, slightly higher in the case without RUWIs. This phenomenon is likely caused by earlier water breakthrough in the case with RUWIs, potentially reaching production constraints (e.g., lower tubing head pressure limits) and resulting in the occurrence of temporary shut-in in some wells. These observations highlight a critical trade-off; while RUWIs enhance pressure and oil recovery, they seem to accelerate water encroachment, potentially limiting well operability. This underscores the importance of a balanced evaluation approach. In other words, simulation results need to be interpreted not solely based on water cut profiles, but in conjunction with operational constraints and flow capacity metrics, to avoid overlooking "hidden" inefficiencies.

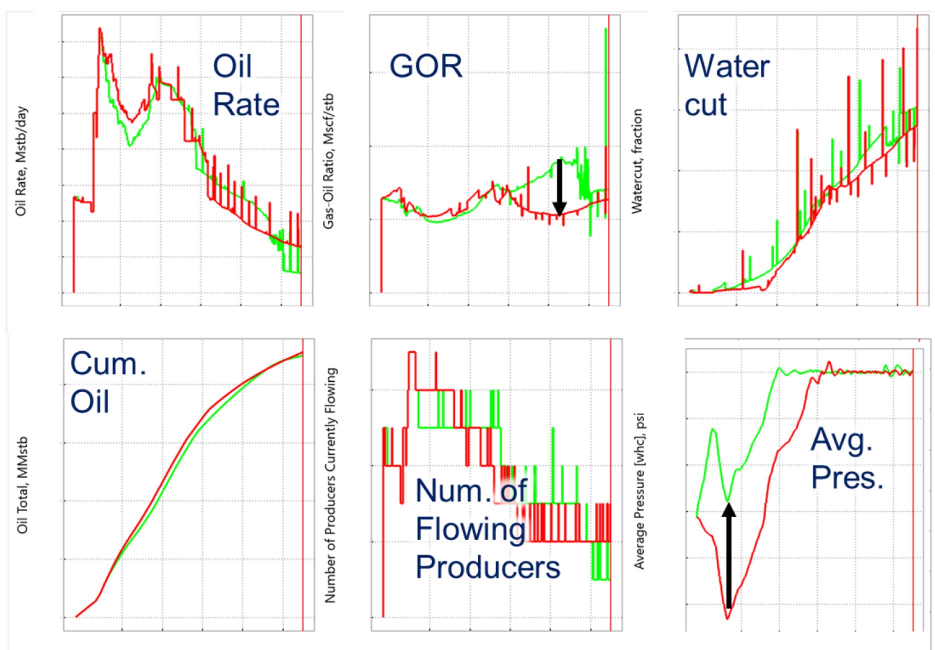

Figure 7—Sector-wise profile comparison of cases with and without RUWIs. Green curves represent the case with active RUWIs and red curves represent the case with inactive RUWIs. Cumulative oil shows an increase in the case with RUWIs, accompanied by the higher average pressure in this sector caused by effective pressure support to the oil pool.

Inner-Ring Water Injectors (IRWI).

Streamline Visualization - IRWI

IRWIs are strategically positioned near the GOC zones. Because currently sanctioned development plan includes only partial coverage with this type of injectors (rather than a full-ring implementation), streamline visualizations were conducted for two scenarios with and without IRWIs to assess their impact (Fig. 8). A notable change in flow patterns is observed when comparing these two scenarios. In particular, streamlines originating from the gas injectors—represented in purple and green—show a significant reduction in westward migration (i.e., toward Sector 3, the left-hand side of the figure) in the presence of IRWIs. Despite the constant gas injection rate in both cases, the flow direction of the injected gas is altered. These variations indicate that IRWIs not only suppress undesired westward gas migration to Sector 3 but also enhance drive toward different sectors, contributing to possible more efficient reservoir management.

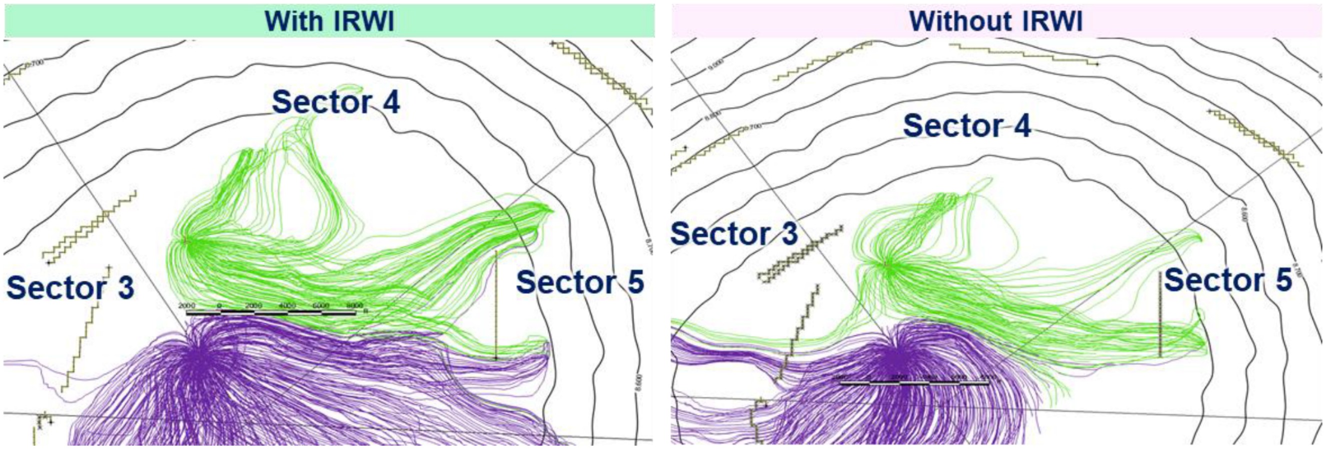

Figure 8—Streamline comparison from gas injectors associated with and without the IRWIs. The presence of IRWI alters the gas flow direction, narrowing the streamlines and indicating a constrained sweep influenced by the inner-ring water injection.

Pressure Map Comparison - IRWI

Pressure distribution comparison was conducted five years after the activation of the IRWIs, to evaluate their contribution to reservoir pressurization. The results reveal a moderate but consistent increase in reservoir pressure within the oil zone when IRWIs are included as illustrated in Fig. 9. The pressure uplift is typically in the range of several tens to approximately 100 psi. In the scenario without IRWIs, higher pressure is expected in the vicinity of them indicating high injection volume to this area. This means some amount of injected volume goes to the IRWI affecting the global allocation of injection water among the injectors. In other words, in the absence of IRWIs, a greater share of the total injection volume is allocated to RUWIs based on well potential, resulting in locally increased pressures. This indicates, again, dynamic reallocation of injection volumes between the different injector categories in particular RUWIs and IRWIs in the current case. Such controlled pressurization may offer operational advantages by mitigating risks associated with premature water breakthrough.

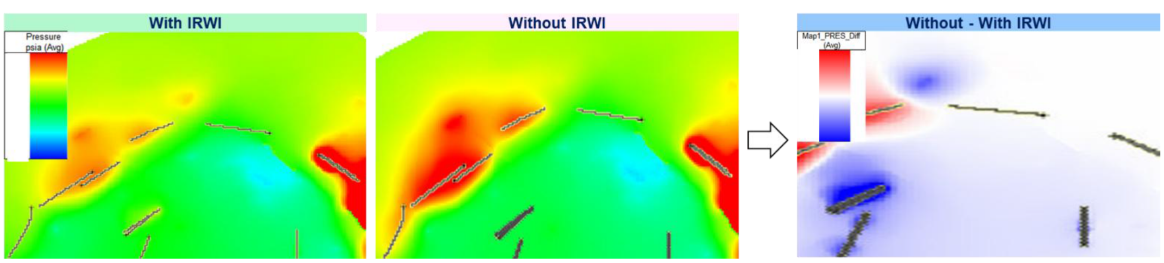

Figure 9—Pressure difference map comparing cases with and without IRWIs. A moderate pressure increase is expected with the IRWIs. This is caused by higher water allocation to IRWIs, which reduces injection through RUWIs. As a result, the pressure difference around RUWIs appears lower.

Simulation Profile - IRWI

Impact of IRWIs on production behavior was evaluated in Sector 3. The comparison between cases with and without IRWIs reveals that differences in cumulative oil production and oil rates are not pronounced as shown in Fig. 10. Nevertheless, a notable reduction in the GOR is observed in the presence of IRWIs, indicating their effectiveness in controlling gas migration. The GOR suppression is primarily caused by the role of the IRWIs as a gas containment mechanism, acting as a barrier for gas expansion and delaying gas breakthrough to production wells. Additionally, a slight decrease in water cut is also observed, which can be attributed to the redistribution of injected water across the injection network. Specifically, in the case with IRWIs, the volume of water allocated to RUWIs decreases, reducing the likelihood of early water breakthrough from those wells. Although the pressure enhancement induced by the IRWIs is moderate—typically around 50 psi—the added support contributes to a more favorable pressure profile throughout the sector (i.e., less localized higher pressure). This, in turn, facilitates stable flowing conditions for a greater number of wells over their production lifespan. These observations suggest that IRWIs contribute both directly and indirectly to improved reservoir management—directly, by lowering GOR and acting as a gas shield and indirectly, by alleviating injection overload on RUWIs and stabilizing pressure distribution. The interplay between injection allocation, GOR suppression, and water cut behavior highlights the sensitive trade-offs involved in injector placement strategies within highly heterogeneous carbonate reservoirs.

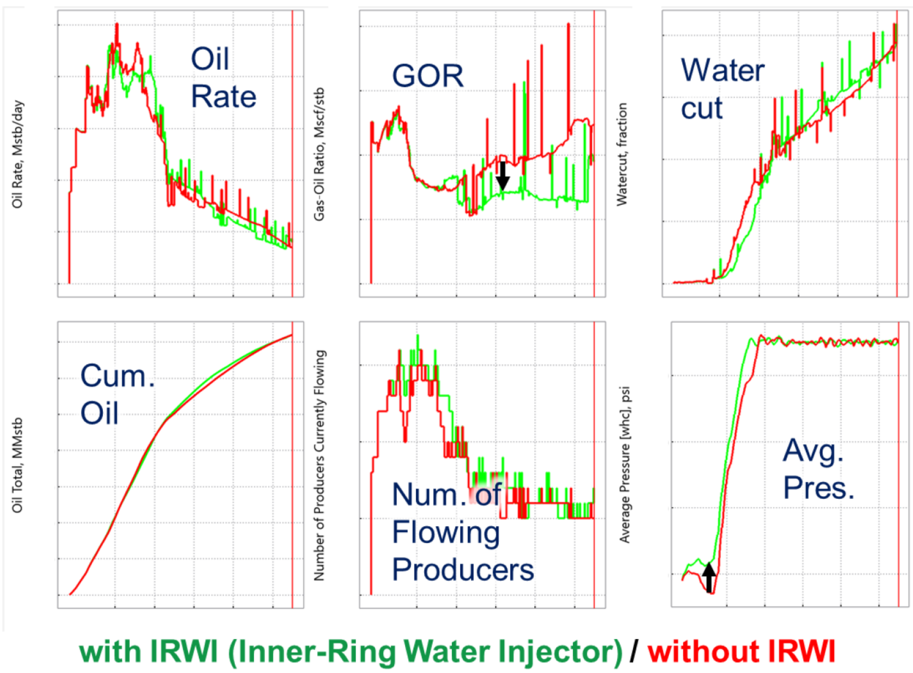

Figure 10—Comparison of sector performance with and without IRWIs. Average pressure shows slight improvement, and GOR decreases significantly when IRWIs are active.

Statistical simulation: Correlation Exploration with Experimental Design

This section explores statistical correlations between injector constraints and sector-level performance using simulation results derived from a LHS-based experimental design. As described in the Methods section, the experimental design included 16 independent variables distributed across six sectors: five DDWI groups, six RUWI groups, and five IRWI groups. Each injector group was assigned a unique upper limit on injection rate within its sector, and the simulation set was constructed accordingly to capture a wide range of operational scenarios.

Down-Dip Water Injectors (DDWI). For the DDWIs, the simulation profiles were plotted in Fig. 11 against oil recovery metrics five years after injection commencement. These scatter plots were examined to identify any underlying correlations between down-dip injection intensity and performance indicators such as cumulative oil production or reservoir pressure. The analysis reveals a consistent lack of significant correlation between down-dip injection rates and sector-level performance. The scatter plots show widespread dispersion, suggesting that whether DDWIs operate near its lower or upper injection rate, its influence on key production indicators remains marginal. This implies that activating or prioritizing DDWIs cannot be strongly justified based on these simulation results alone.

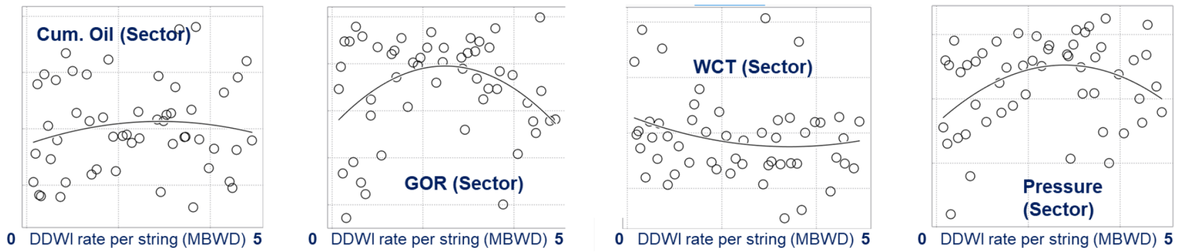

Figure 11—Statistical simulation results showing the correlation between DDWI rates and sector-level performance. 55 cases were generated using LHS. Cumulative oil, GOR, water cut, and average pressure in a target sector are plotted against the average injection rate per DDWI string.

However, this conclusion should not be interpreted as a definitive recommendation against down-dip injection at this stage. Rather, it highlights the necessity for complementary justifications by engineering judgement that may validate the deployment of DDWIs beyond the statistical behavior captured in this experimental design.

Roll-Up Water Injectors (RUWI). The RUWIs exhibited a distinct set of correlations with key reservoir performance metrics, offering valuable insights into the interplay between injection intensity and sector-level outcomes. Scatter plots were generated as shown in Fig. 12 for each sector by comparing the injection rates against simulation results estimated five years after the commencement of injection. The cumulative oil production initially remains stable across a wide range of injection rates (from 0 to approximately 5 MSTB/D). However, beyond this threshold, a noticeable decline in cumulative oil is observed as injection rates increase toward 10 MSTB/D, suggesting diminishing returns or potentially negative impacts at high injection levels. This trend implies that excessive injections from RUWIs may lead to suboptimal recovery, likely caused by premature water breakthrough. GOR also demonstrates a declining trend with increasing injection rates, leveling off at higher injection volumes. This suggests that increased pressure support from RUWI is effective at suppressing high GOR. Water cut, on the other hand, shows a continuously increasing trend with injection intensity, indicating accelerated water breakthrough and increasing water production challenges. Average reservoir pressure exhibits a mild upward trend with increasing injection rates. However, the magnitude of pressure buildup is relatively limited. This behavior may be attributed to the pressure drawdown caused by aggressive oil production to meet oil rate targets, which offsets the pressure gains achieved via water injection.

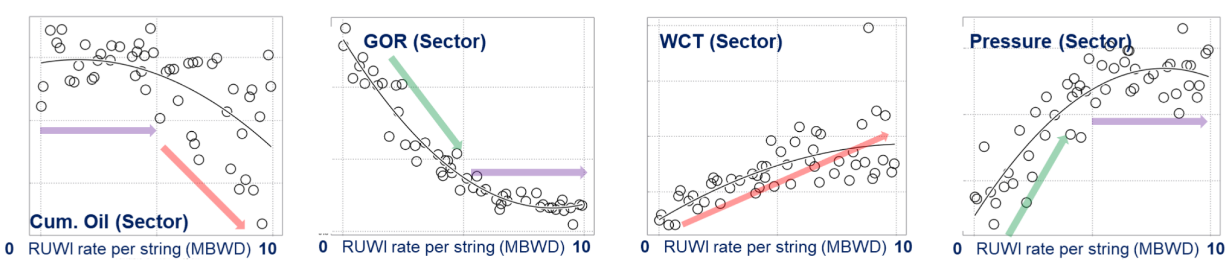

Figure 12—Statistical simulation results showing the correlation between RUWI rates and sector-level performance. 55 cases were generated using Latin Hypercube Sampling. Cumulative oil, GOR, water cut, and average pressure in a target sector are plotted against the average injection rate per RUWI string.

These findings highlight the existence of complex trade-offs among key metrics: while RUWIs effectively enhance pressure maintenance and reduce GOR, they also tend to accelerate water production and potentially reduce oil recovery at high injection volumes. The results highlight the importance of calibrating injection strategies to balance pressure support, gas containment, and water management, rather than maximizing injection volumes indiscriminately.

Inner-Ring Water Injectors (IRWI). The injection behavior of IRWIs exhibits relatively consistent and interpretable trends across sectors, despite some local variability as shown in Fig. 13. Increasing the upper-limit injection rates generally yields a positive impact on cumulative oil production. This trend supports the hypothesis that IRWIs, when placed strategically near the gas-oil contact, enhance reservoir sweep efficiency and delay adverse gas-related effects. As expected, the GOR shows a decreasing trend with increasing injection volume, indicating that early gas breakthrough is effectively mitigated. Average reservoir pressure, however, does not exhibit a strong or consistent correlation with IRWI rates, likely caused by the distributed nature of the pressure support provided by these wells and the competing effects from other injection categories. Interestingly, water cut displays a slight downward trend as inner-ring injection rates increase. While this might appear counterintuitive—since more water injection often correlates with higher water production —this behavior can be interpreted in the broader context of injection allocation. Specifically, as more injection volume is directed to IRWIs, the injection rate allocation to RUWIs is reduced. This, in turn, helps suppress rapid water advancement typically observed from roll-up wells, resulting in a more balanced fluid front and a delayed water breakthrough in nearby producers.

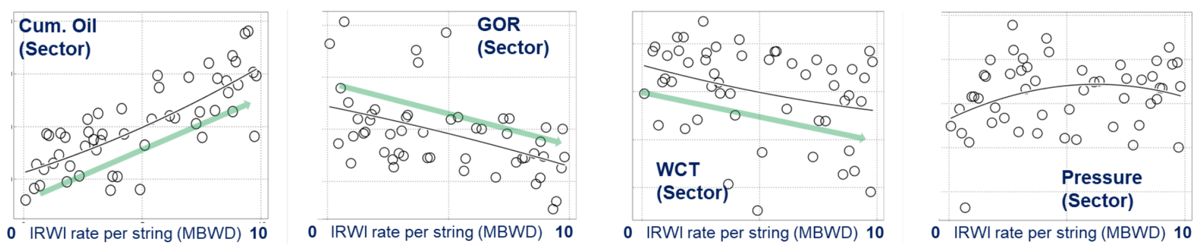

Figure 13—Statistical simulation results showing the correlation between IRWI rates and sector-level performance. 55 cases were generated using Latin Hypercube Sampling. Cumulative oil, GOR, water cut, and average pressure in a target sector are plotted against the average injection rate per IRWI string.

These observed trends, although not universally monotonic across all sectors, provide meaningful statistical guidance for designing robust injection strategies. The results suggest that IRWIs not only contribute to oil recovery but also play a regulatory role in systemwide injection dynamics by mitigating undesirable flow behavior from RUWIs.

Spearman Correlation. In the final part of the statistical simulation analysis, Spearman correlation evaluation was conducted using 55 realizations derived from LHS as summarized in Table 2. Sixteen independent injection variables, representing upper-limit injection rates across six sectors, were compared against four key performance indicators: cumulative oil production, GOR, water cut, and average pressure. Unlike Pearson correlation, Spearman correlation—based on ranked values—offers more robust insights into non-linear, non-monotonic relationships, which are common in reservoir systems.

Table 2—Spearman correlation values are coefficients between injector-specific rates and sector-level performance indicators. Four correlation matrices are presented for cumulative oil production, GOR, water cut, and average pressure in each sector (1-6). Each table lists correlation values between injection rates of DDWI, RUWI, and IRWI (grouped by sector 1-6) and the corresponding sectoral performance metrics. High values represent high correlation whether negative or positive.

Cumulative Oil Production

		Sector 1	Sector 2	Sector 3	Sector 4	Sector 5	Sector 6
DDWI	1	0.23	0.56	0.27	0.10	0.45	0.13
	2	0.07	0.15	0.32	0.12	0.11	0.16
	3	-	-	-	-	-	-
	4	0.07	0.14	0.32	0.55	0.34	0.08
	5	0.15	0.15	0.27	0.06	0.11	0.20
	6	0.18	0.22	0.33	0.16	0.28	0.12
RUWI	1	0.91	0.08	0.16	0.20	0.26	0.21
	2	0.20	0.52	0.28	0.25	0.11	0.08
	3	0.20	0.38	0.77	0.31	0.10	0.18
	4	0.21	0.17	0.40	0.54	0.29	0.50
	5	0.31	0.27	0.18	0.28	0.96	0.27
	6	0.10	0.17	0.08	0.18	0.28	0.22
IRWI	1	0.81	0.28	0.27	0.25	0.37	0.31
	2	0.26	0.29	0.23	0.25	0.11	0.22
	3	0.23	0.23	0.61	0.31	0.14	0.30
	4	0.22	0.29	0.51	0.17	0.30	0.22
	5	0.18	0.26	0.27	0.43	0.50	0.22
	6	-	-	-	-	-	-

GOR

		Sector 1	Sector 2	Sector 3	Sector 4	Sector 5	Sector 6
DDWI	1	0.17	0.36	0.42	0.25	0.43	0.37
	2	0.09	0.25	0.28	0.17	0.15	0.14
	3	-	-	-	-	-	-
	4	0.13	0.32	0.25	0.34	0.35	0.39
	5	0.20	0.31	0.21	0.23	0.16	0.19
	6	0.22	0.19	0.28	0.28	0.15	0.19
RUWI	1	0.78	0.30	0.16	0.19	0.32	0.45
	2	0.23	0.47	0.36	0.24	0.19	0.16
	3	0.21	0.96	0.98	0.40	0.21	0.31
	4	0.37	0.22	0.39	0.85	0.27	0.27
	5	0.29	0.31	0.07	0.54	0.97	0.08
	6	0.11	0.27	0.08	0.27	0.20	0.74
IRWI	1	0.84	0.42	0.36	0.34	0.34	0.29
	2	0.44	0.26	0.25	0.27	0.23	0.12
	3	0.39	0.36	0.50	0.24	0.24	0.16
	4	0.14	0.17	0.32	0.33	0.27	0.09
	5	0.25	0.20	0.13	0.43	0.63	0.30
	6	-	-	-	-	-	-

Water Cut

		Sector 1	Sector 2	Sector 3	Sector 4	Sector 5	Sector 6
DDWI	1	0.18	0.39	0.38	0.36	0.30	0.38
	2	0.25	0.22	0.34	0.24	0.24	0.24
	3	-	-	-	-	-	-
	4	0.37	0.33	0.34	0.63	0.29	0.29
	5	0.13	0.22	0.28	0.12	0.08	0.25
	6	0.12	0.26	0.27	0.22	0.15	0.28
RUWI	1	0.79	0.17	0.31	0.26	0.17	0.28
	2	0.34	0.52	0.28	0.33	0.13	0.24
	3	0.18	0.75	0.97	0.25	0.22	0.19
	4	0.13	0.22	0.23	0.45	0.37	0.32
	5	0.34	0.38	0.25	0.73	0.94	0.25
	6	0.19	0.17	0.13	0.28	0.33	0.61
IRWI	1	0.66	0.74	0.33	0.28	0.32	0.25
	2	0.60	0.28	0.14	0.17	0.14	0.24
	3	0.44	0.46	0.59	0.43	0.19	0.22
	4	0.25	0.18	0.40	0.26	0.12	0.34
	5	0.40	0.28	0.17	0.28	0.81	0.24
	6	-	-	-	-	-	-

Pressure

		Sector 1	Sector 2	Sector 3	Sector 4	Sector 5	Sector 6
DDWI	1	0.25	0.10	0.23	0.19	0.15	0.37
	2	0.08	0.32	0.19	0.04	0.10	0.16
	3	-	-	-	-	-	-
	4	0.10	0.16	0.22	0.27	0.49	0.46
	5	0.11	0.12	0.18	0.04	0.10	0.18
	6	0.05	0.16	0.35	0.23	0.22	0.18
RUWI	1	0.96	0.13	0.19	0.15	0.33	0.20
	2	0.07	0.69	0.15	0.25	0.20	0.10
	3	0.38	0.10	0.91	0.37	0.14	0.21
	4	0.13	0.14	0.29	0.85	0.37	0.13
	5	0.10	0.16	0.18	0.09	0.96	0.11
	6	0.04	0.13	0.10	0.16	0.19	0.81
IRWI	1	0.23	0.37	0.25	0.34	0.47	0.23
	2	0.18	0.21	0.11	0.16	0.24	0.12
	3	0.13	0.33	0.27	0.21	0.09	0.12
	4	0.21	0.34	0.30	0.33	0.09	0.09
	5	0.26	0.19	0.12	0.30	0.64	0.10
	6	-	-	-	-	-	-

Cumulative Oil Production Correlation

Among all injector types, RUWIs exhibited the strongest positive correlation with cumulative oil production across multiple sectors. The correlation matrix revealed that injector variables generally influence the performance of the sectors in which they are located, as seen through strong diagonal trends in the table. For example, injectors assigned to Sector 1 showed a clear influence on oil recovery within the same sector. IRWIs also demonstrated notable positive correlations, particularly in Sectors 1 and 3, while down-dip injectors showed relatively weak or inconsistent associations.

GOR Correlation

Clear and consistent negative correlations were observed between RUWI variables and GOR across all sectors, indicating their role in pressure maintenance and gas management. IRWIs, especially those affecting Sector 1, also contributed to GOR suppression. Interestingly, in some cases, injectors located in one sector had notable influence on adjacent sectors, underscoring the cross-boundary effects of fluid migration and the artificial nature of sector divisions, which do not have any geological barriers.

Water Cut Correlation

RUWIs again showed the strongest correlation with water cut, though the directionality (positive or negative) requires contextual interpretation through supporting plots. High injection from RUWIs tends to accelerate water breakthrough, leading to a potential increase in water cut. Conversely, IRWIs help mitigate this effect by redistributing injection volumes, thus indirectly reducing water production by lowering flow into roll-up wells. This finding supports the concept of strategic reallocation of total field water injection to emphasize IRWIs without increasing the overall VRR.

Pressure Correlation

Sector-level average pressure, particularly hydrocarbon-weighted pressure, was most strongly influenced by RUWIs. These wells, situated near the oil zone, have a direct impact on reservoir pressure, which is critical for maintaining productivity. This statistically validates the operational observation that RUWIs are highly effective in enhancing phase pressure recovery, especially in oil-dominated layers.

Trial of Neural Network Modeling. As part of the statistical simulation efforts, a neural network-based proxy model was developed using the 55 Latin Hypercube simulation cases. Although the model demonstrated a high degree of accuracy in fitting the training data, it exhibited significant overfitting. Even when a subset of 40 cases was used for training, the model failed to provide reliable predictions for the remaining 15 validation cases, indicating poor generalizability. This outcome suggests that the problem space is governed by complex, nonlinear interactions that are not easily captured by a single proxy model. While it is technically feasible to build a model that fits the available data well, such a model risks being misleading and potentially detrimental if interpreted without caution.

The attempt supported a key insight: purely data-driven approaches may not always provide meaningful insights in reservoir development planning, particularly in systems with intricate dynamic behaviors. This trial served as a strong motivation to incorporate engineering judgment into the workflow. Instead of relying solely on AI models, the integration of deterministic analysis, statistical exploration, and domain expertise was recognized as essential for robust decision-making. Hence, the neural network model was not adopted for the final optimization process, but the lessons learned reinforced the importance of expert-guided interpretation when dealing with limited or high-dimensional datasets.

Engineering Judgement: Expert-led Analytical Evaluation

From a macro perspective, if the judgmental interpretation is reasonably consistent with deterministic and statistical simulation outputs, it can validate those simulations and support their use in field-level decision-making. Thus, engineering judgment does not oppose modeling—it complements and validates it through critical cross-checking. This is the objective in which we present the following observations.

Down-Dip Water Injectors. This paragraph uses the southern sector, Sector 1, to show what DDWI actually delivered at field scale. Fig. 14 brings together the southern sector's injection and production history with the producer BHP. During the period of continued down-dip water injection, producer BHP kept trending downward, and oil rate changed little through the early and middle stages. This points to pressure support that did not propagate into the producing fairway. When crestal gas injection started, the BHP curve flattened and became steadier, although the absolute level remained modest. The contrast indicates that, in this geometry, crest control stabilized the pressure field more effectively than additional downdip water. For this sector, DDWI shows low leverage on near-term oil response and on hydrocarbon-weighted pressure. With a capped total WI budget, further downdip water should be deprioritized in favor of up-dip levers that act closer to the producing pay. DDWI can be kept locally for edge containment/voidage balance where there is a clear need, but it should not displace up-dip allocation. This conclusion is consistent with the record in Fig. 13: more DDWI without BHP recovery, little change in oil rate in the early/mid period, and BHP stiffening coincident with the start of gas injection.

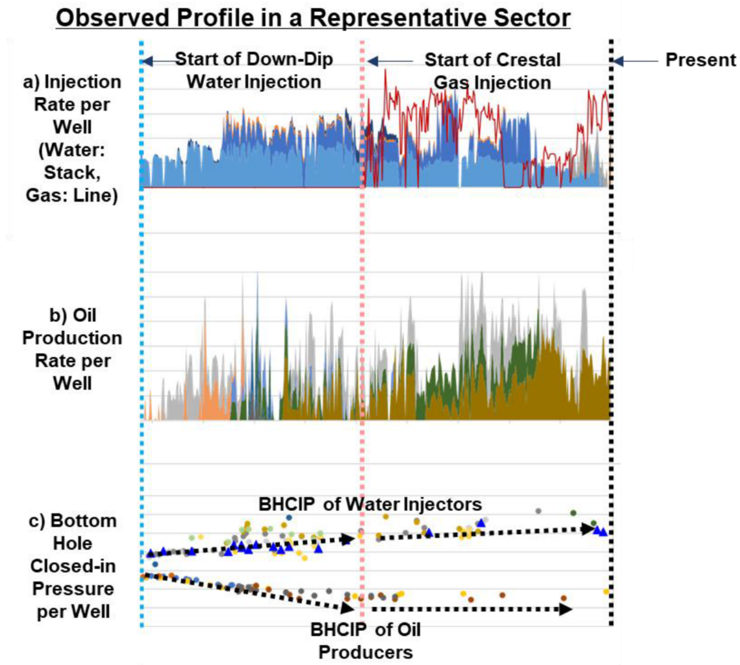

Figure 14—Southern Sector history profile: (a) Injection rate: stack for water injection (DDWI) and line for crestal gas injection.(b) Oil rate, (c) Bottom Hole Closed-in Pressure — declining through the DDWI phase; trend flattens following GI onset, indicating stronger crest-driven pressure stabilization than from downdip water.

Roll-Up Water Injectors & Inner-Ring Water Injectors. The north-west sector, Sector 3, is high-GOR and operates at comparatively high pressure. In this area, the highest-GOR producer sits between a RUWI and an IRWI. Fig. 15 shows the producer profile after the pair was brought online. Following RUWI start-up, a prompt drop in GOR was recorded. With the IRWI active, the lower GOR has persisted and the well has remained stable. Oil rate has been steady, and there is no early rise in water cut or sign of breakthrough. Injectivity and day-to-day operations are normal.

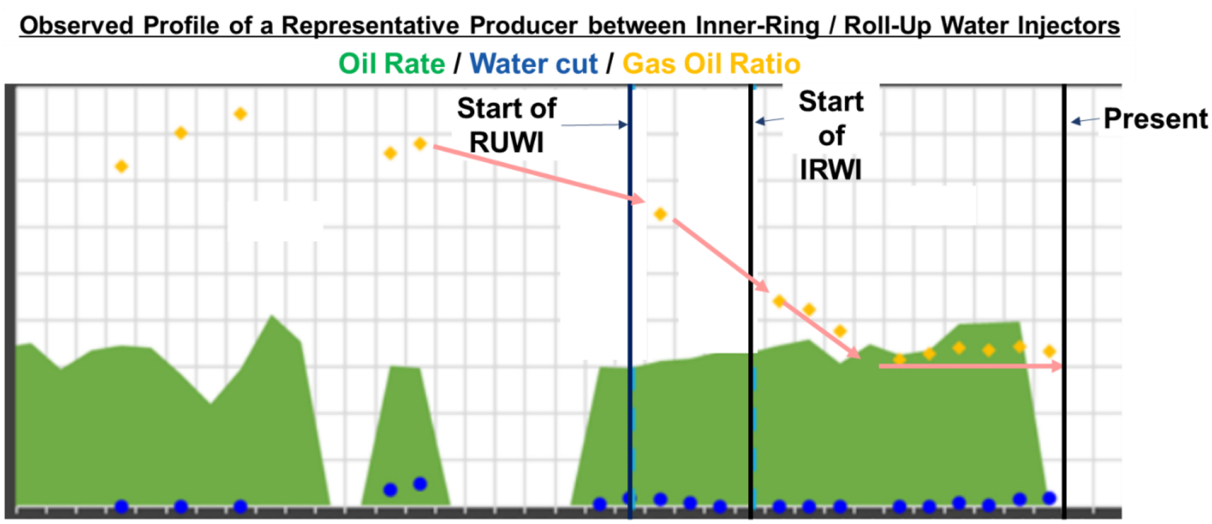

Figure 15—Producer between RUWI and IRWI: timeline of injector activation and producer response. GOR drops promptly after RUWI starts and remains lower with IRWI active; oil rate stable; no early water cut increase.

This paired pattern is performing as intended: RUWI provides pressure support while IRWI limits gas movement towards oil zone, allowing the producer to run with lower GOR and without added water risk so far. Because the pair has been in operation only a short time, the pressure-recovery signal will take longer to confirm than the GOR response. For the robust operation, it is recommendable to continue surveillance —trend sector pressure and producer BHP, track GOR and pattern-level water cut, and plan follow-up flow tests and targeted production logs, and tracer or interference checks if practical.

Integration of Analysis of Deterministic / Statistical Simulations and Engineering Judgement

An integrated interpretation of the deterministic simulations, statistical correlation analyses, and engineering evaluations is summarized in Table 3. While specific sector-by-sector injection volumes are not disclosed due to confidentiality, the table presents general trends that inform strategic planning and injector allocation across the reservoir.

Table 3—General summary of injector types with associated pros and cons.

DDWI (Down-Dip Water Injector)	RUWI (Roll-Up Water Injector)	IRWI (Inner-Ring Water Injector)
- DDWI would have **a minor benefit** on RUWI efficiency up to 10%. - Due to WI rate upper limits, **WI volume allocation** from RUWI & IRWI to DDWI **might delay the reservoir pressure restoration** in oil zone after 5 – 10 years of injection.	- RUWI is a key driver to **restore the reservoir pressure**. - However, RUWI requires **sensitive control** to avoid reducing oil production due to **early water BT**.	- IRWI plays a key role to **improve oil production by reducing GOR** as water fence and increasing pressure. - IRWI in some sectors would help the gas migrate into the other sectors leading to high GOR, creating a **sensitive trade-off** between **in-situ-gas-lift benefit** and **GOR constraints**.

Down-Dip Water Injectors (DDWI). The role of DDWIs in contributing to overall reservoir support is acknowledged, particularly in facilitating pressure propagation toward structurally higher areas. However, the incremental impact of DDWIs—compared to the absence of DDWIs—remains modest, with water injection efficiency improvements of RUWI typically limited to around 10%pt. Moreover, allocating injection capacity to these wells inherently reduces the available volume for RUWIs or IRWIs, potentially delaying pressure recovery in more critical oil-bearing zones. This trade-off should be factored into any full-field injection strategy.

Roll-Up Water Injectors (RUWI). RUWIs are recognized as a primary driver for restoring reservoir pressure, especially in up-dip regions. Their deployment significantly aids in maintaining pressure above the bubble point and enhancing oil mobility. Nevertheless, when injected at excessive rates, they have shown a tendency to trigger premature water breakthrough, depending on sector-specific geological and flow characteristics. Adaptive rate control, informed by real-time monitoring and sector-level behavior, is essential to mitigate these risks.

Inner-Ring Water Injectors (IRWI). The function of IRWIs as an effective water-fence mechanism is clearly demonstrated. Increased injection through these wells tends to reduce gas breakthrough, thereby lowering GOR and extending oil production under constrained operating envelopes. This GOR suppression contributes to improved recovery efficiency. However, such effects are not without trade-offs. In zones where gas-lift efficiency is beneficial for production, GOR reduction may impede natural lift performance. Conversely, in unconstrained sectors, redirected gas migration resulting from altered injection distributions may enhance lift and production. These inter-sector interactions reinforce the necessity of dynamic injection control based on ongoing reservoir diagnostics.

This integration illustrates that effective reservoir management demands more than isolated reliance on deterministic forecasts or statistical inferences. Simulation artifacts and production-system interactions introduce layers of complexity that must be navigated through engineering judgment. A flexible, feedback-informed injection strategy—calibrated through simulation insights and updated through production surveillance—offers the most robust path forward for maximizing recovery while minimizing operational risks.

Conclusions

This study presented an integrated injection strategy optimization for a heterogeneous carbonate reservoir with over 140 water injectors including history. By combining deterministic simulation, statistical correlation analysis, and engineering judgment, key insights were obtained for different injector categories—Down-Dip, Roll-Up, and Inner-Ring Water Injectors.

- Down-Dip Water Injectors offered limited incremental benefits and, in certain cases, delayed reservoir pressure restoration due to their competition with other injectors for available injection volume.
- Roll-Up Water Injectors were confirmed to be primary drivers of reservoir pressure support; however, excessive injection rates led to rapid water breakthrough, emphasizing the need for sector-specific rate control.
- Inner-Ring Water Injectors effectively acted as gas control barriers, lowering GOR and improving oil recovery efficiency. Nevertheless, over-injection could unintentionally suppress autonomous gas-lift mechanisms in neighboring sectors.

The optimal design must therefore be dynamic and flexible—capable of adapting to reservoir feedback while balancing injection distribution. Engineering judgment remains essential in interpreting model sensitivities and ensuring robust development planning beyond simulation artifacts.

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